problem 1 proof

definitions

let S be the number of stones ($1 \leq S \leq 10^{10^5}$). a "move" is to subtract a **palindrome** p where $p \in \mathbb{N}$.

terms

- **palindrome**: a number that reads the same backwards and forwards, but **leading zeros are not allowed**.
- **winning position**: has a number of stones where the current player can make a move that forces a win
- **losing position**: no matter what move the current player makes, the other player will be in a **winning position**.

relationships

1. a position is a **winning position** if there exists at least one valid move to a **losing position**.
2. a position is a **losing position** if all valid moves lead to **winning position**.
3. when $S = 0$, it is a **losing position**.

claim

theorem

for any number of stones $S > 0$:

$$S \in \text{winning} \iff S \not\equiv 0 \pmod{10}$$

$$S \in \text{losing} \iff S \equiv 0 \pmod{10}$$

lemma any natural number **palindrome** p is not divisible by 10.

$$\forall p \in \mathbb{N}, \text{palindrome}(p) \Rightarrow p \not\equiv 0 \pmod{10}$$

proof of lemma: a number is divisible by 10 if and only if its last digit is 0. for a number p to be a **palindrome**, its string of digits must read the same forwards and backwards; and according to that definition, if the last digit is 0, the first digit must also be 0. however, in the definition, it states that leading zeros are not allowed. therefore, a positive **palindrome** cannot have 0 as its first digit, and thus cannot have 0 as its last digit. this means p cannot be a multiple of 10. $\blacksquare$

proof of theorem

proof using *strong induction* on S.

base case checking small values:

- if $S \in \{1, 2, 3, \ldots, 9\}$, the player can remove all $p \equiv S$ stones, since any single digit is a **palindrome**.
- if $S = 10$, the player cannot remove all S stones, since it is not a **palindrome** (by our lemma). they will have to remove p stones so that $S = 10 - p$, where $S \in \{1, 2, 3, \ldots, 9\}$. according to our lemma and the case above, the next player can remove all $p \equiv S$ stones, putting the current player at a **losing position**.
- if $S < 10$, the player can remove all S stones, since it is a valid **palindrome**, putting them in a **winning position**.

inductive hypothesis assume that for all integers k such that $0 < k < S$, our theorem holds – where k is a **losing position** *if and only if* $k \equiv 0 \pmod{10}$.

inductive step we consider two cases:

case 1: $S \equiv 0 \pmod{10}$ we want to show that any move to S is a **losing position**, so we must show that any valid move from S leads to a **winning position**.

let the current player remove p stones, where p is a **palindrome** and $p \leq S$. the new number of stones are now $S' = S - p$.

we know that:

1. $S \equiv 0 \pmod{10}$ by the case definition; and
2. $p \not\equiv 0 \pmod{10}$ by our lemma.

so the new number of stones $S' \pmod{10}$ is:

$$\begin{aligned} S' \pmod{10} &= (S - p) \pmod{10} \\ &= (0 - (p \pmod{10})) \pmod{10} \quad \text{(substitution)} \end{aligned}$$

since $p \pmod{10} \in \{1, 2, 3, \ldots, 9\}$, it follows that $S' \pmod{10} \neq 0$. therefore, any move from S results in a new pile S' which is not a multiple of 10. since $p > 0$, we have $S' < S$.

by our inductive hypothesis, any such S' is a **winning position**. since every possible move from S leads to a **losing position**, S must be a **losing position**.

case 2: $S \not\equiv 0 \pmod{10}$ we want to show that some move from S leads to a **winning position**, so we must show there exists *at least one* move from S to a **losing position**

assume the player chooses to move p stones, where $p = S \pmod{10}$. the move should satisfy these requirements:

- since $S \not\equiv 0 \pmod{10}$, p will be an integer from 1 to 9
- $p \in \{1, 2, 3, \ldots, 9\}$, which is a **palindrome** according to our definition
- $p = S \pmod{10} \leq 9 < S$ for $S > 10$. if $S < 10$, $p = S$ is also a valid move.

the new number of stones will be:

$$\begin{aligned} S' &= S - p \\ &= S - (S \pmod{10}) \quad \text{(substitution)} \end{aligned}$$

by that definition, S' is now also divisible by 10.

$$\begin{aligned} S' &= S - (S \pmod{10}) \\ S' \pmod{10} &= S - (S \pmod{10}) \pmod{10} \\ (S \pmod{10}) - (S \pmod{10})) &= 0 \pmod{10} \qquad \text{(subtraction)} \end{aligned}$$

since $p > 0$, we have $S' < S$. by our inductive hypothesis, this new position S' is a **losing position**.

we have found a move from S to a **losing position**, which means S is an **winning position**. this proves the claim for this case.

conclusion

both cases hold, so by the principle of strong induction, our theorem is true for all natural numbers.

the first player wins if the starting number of stones S is not a multiple of 10. the next player wins if S is a multiple of 10.

since a number ending in 0 is exactly when $S \pmod{10} = 0$, the winning strategy is simply to check if the last digit of S is 0. ■